

ANUBHAV AGRAWAL

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SUMMARY

Machine Learning Engineer with **3+ years** owning AI systems end-to-end — model architecture, real-time inference, *and* the production platform around them. Shipped a **\$200K+ ARR** computer-vision product line (**600+ FPS, 3 patents**), built the retrieval and evaluation core of an LLM product serving **500+ users** plus its AWS deployment, **CI/CD**, and **Grafana LGTM** observability. Author of open-source **LLM-serving benchmarks** quantifying when local inference stops scaling.

PROFESSIONAL EXPERIENCE

Griphic

Delhi

AI Engineer - 2

Jan 2026 – Present

- Engineered the **adaptive-intelligence core**: a stateful **LangGraph** agent that scores responses, scales **difficulty live**, and retrieval-conditions follow-ups into an interviewer-grade assessment.
- Architected the **RAG pipeline** with **3 parallel retrieval paths** (conversation embeddings, MongoDB, knowledge vault), *hybrid keyword + dense search* and *cross-encoder re-ranking*: **0.95 Recall@5** over **10K+ docs** with **512-d** embeddings.
- Cut query latency **2.5s → 0.8s (68%)** via context-fusion & async orchestration; **+42%** domain-query **F1** on a held-out eval set; serves **500+ users** at **99.9% uptime**.
- Built the production system end-to-end — **Python/FastAPI** on **AWS EC2** with **MongoDB** and **OpenAI APIs** for LLM inference — and designed its **CI/CD pipeline** with Docker image builds.
- Stood up observability: **OpenTelemetry Collector** shipping logs, traces, metrics to **Grafana LGTM (Loki, Tempo, Mimir)**; built **LLM token-tracking and cost-estimation** dashboards for per-feature cost visibility.
- Built an **eval harness** (recall, faithfulness, answer-relevance) and citation-grounded **LLM guardrails** for safe outputs.

Innvolution Healthcare Pvt. Ltd.

Bangalore

Machine Learning Engineer & Team Lead

Sep 2023 – Dec 2025

- Owned **AIMAG** — an X-ray super-resolution & denoising product at **\$200K+ ARR** — through model R&D, quantization, and release; built its serving stack (**PyTorch → ONNX → TensorRT, INT8/FP16, FastAPI/C++**): **600+ FPS** on RTX 4090, **70% latency cut**, **<5% quality loss**, **35% lower VRAM**.
- Built an automatic **QCA** pipeline (segmentation → calibration → prediction) in **<4 s** on a fine-tuned **U-Net++ (ResNet50 + attention)** at **0.94 Dice**.
- Led a **6-engineer ML team**, owning roadmap, review standards, and the lifecycle through **clinical validation**.

Image Quality Engineer Intern

Nov 2022 – May 2023

- Built motion-compensated stitching & denoising for **DSA** sequences — **20% clarity gain** at **<2-pixel** alignment error via *optical-flow* registration.

SELECTED PROJECTS

Inference Lab — LLM Serving Benchmarks on Apple Silicon

[GitHub](#)

Reproducible benchmark suite: mlx-lm vs llama.cpp under one fairness contract

2026

- Built the harness — fixed 12-prompt workload, greedy decode, p50/p99 **TTFT**, **TPOT**, aggregate throughput, peak RSS — and measured **45–56 tok/s** single-stream on a 4-bit 7B model.
- Proved single-instance throughput is **flat from 1 to 32 concurrent users**, then built a K-instance process-isolated dispatcher showing **+9% for 3× the memory**: one instance already saturates **~250 of the M4 Pro's ~273 GB/s** memory bandwidth. Derived the local-vs-cloud decision boundary; next phase benchmarks continuous batching (**vLLM**, **TensorRT-LLM**) on cloud GPUs.

EDUCATION

Birla Institute of Technology & Science (BITS), Pilani — M.Tech in AI & Machine Learning

2025 – 2027

Bhilai Institute of Technology, Durg — B.Tech in Computer Science Engineering

2019 – 2023

TECHNICAL SKILLS

Languages & Tools: Python, C++, Git, Linux, $\text{\LaTeX}$

ML & Deep Learning: PyTorch, TensorFlow, OpenCV, Scikit-Learn, XGBoost, NumPy, Pandas

LLMs & RAG: OpenAI API, LangGraph, HuggingFace, Vector Search, Cross-Encoder Re-ranking, MongoDB

Inference & Serving: TensorRT, ONNX, INT8/FP16 Quantization, CUDA, FastAPI, mlx-lm, llama.cpp

Platform & MLOps: AWS (EC2), Docker, CI/CD, OpenTelemetry, Grafana LGTM (Loki, Tempo, Mimir)

PATENTS APPLIED

AI-POWERED X-RAY IMAGE ENHANCEMENT (202541071194) — single-image blind super-resolution & denoising. **ENHANCING BODY PART FROM ANGIOGRAPHY LOOP** (202441009957) — dynamic roadmap generation from DSA sequences. **IMAGE ENHANCEMENT IN X-RAY FLUOROSCOPY** (202541041511) — real-time contrast enhancement & denoising.